

Prototype Context-Aware On-Skin Interfaces for Socially Adaptive Biofeedback

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Abstract

Ubiquitous computing opens new possibilities for communicating physiological states through on-skin interfaces, supporting emotional awareness, empathy, and well-being. However, research efforts remain largely exclusive due to barriers in prototyping and a lack of interdisciplinary engagement. In this work, I argue that exploring socially expressive on-skin interaction requires platforms that make prototyping accessible across disciplines. I introduced SkinKit, the first on-skin prototyping toolkit that lowers the entry barrier for making. I developed SkinLink, a fabrication approach that supports on-body adjustments to optimize prototype iteration. I demonstrated ECSkin, an adaptive on-skin display technology of modular electrochromic films, to expand the display modalities. These platforms pave the way for empowering co-design collaborations with multidisciplinary practitioners. To carry forward the explorations, I initiate the research to engage psychology, communication, and design experts to identify design opportunities and considerations for social biofeedback and on-skin interactions. Finally, I describe how achieving context-aware on-skin biofeedback systems can be essential for tailoring feedback to social cues, communication intent, and user-defined preferences. Through this research agenda, I aim to advance a vision of ubiquitous computing that supports interdisciplinary exploration and broadens the context-responsive applications of on-skin interfaces across real-world settings.

CCS Concepts

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing**; **Ubiquitous and mobile computing systems and tools**.

Keywords

On-skin Interface, Prototyping Toolkits, Biofeedback, Participatory Design

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1 problem statement

Physiological computing has become an emerging interaction paradigm that enhances interpersonal understanding by sharing internal psychophysiological signals in social contexts [34]. However, most existing systems rely on detached, screen-based displays [9, 28] that overlook the nuances of social settings and diminish bodily engagement. On-skin interfaces offer a promising alternative: transforming the skin into an expressive medium that externalizes internal states through subtle, embodied cues. This shift reframes the body from a passive sensor to an active participant in communication, enabling new forms of emotional awareness, self-expression, and socially situated meaning-making. While designing on-skin biofeedback interfaces is timely, with recent advancements in epidermal electronics [18, 47] and machine learning approaches for decoding physiological signals in real time [1, 46], realizing these systems' potential requires more than overcoming engineering challenges. It necessitates considering the intricate biological, ethical, and social dimensions related to the intimate interaction between computing systems, physiological signals, and the human body [13].

The complexity of designing socially meaningful on-skin applications calls for multidisciplinary participation in developing biofeedback designs, which resonates with Yvonne Rogers' [37] reappraisal of Marc Weiser's vision of Ubiquitous Computing [44]. Rogers advocates for a shift toward *engaging experiences* that involves and empowers non-ubiquitous practitioners to integrate personal, cognitive, and social processes to contribute to the development of technologies. To approach the vision of embodied and socially responsive biofeedback interfaces, I present the following thesis statement:

To encourage explorations and collaborations of embodied social biofeedback, we should develop accessible prototyping toolkits and engage diverse practitioners to experiment with the on-skin form factor.

In this work, I illustrate a research roadmap that shows the research-through-design process [49] toward socially and emotionally responsive on-skin interfaces. First, I will introduce a series of modular platforms—SkinKit [21], SkinLink [19], and ECSkin [20]—that support low-barrier, reconfigurable on-skin prototyping, enabling participatory collaborations with non-Ubiquitous practitioners. Then, I will describe our work-in-progress study of prototype-led co-design [4] with psychology, communication, and design experts to explore the nuanced experience of biofeedback sharing. Finally, I will reflect on the needs for context-aware on-skin biofeedback systems as the future work. Overall, my work contributes pragmatic toolkits, co-design research methods, and empirical understandings of shared biofeedback to develop socially intelligent on-skin interfaces that adapt to versatile daily-life scenarios.

2 Related Work

2.1 Physiological Computing for Social Interactions

Computing systems with real-time biosignal sensing allow users to reflect on and potentially regulate their internal states, known as the biofeedback technique [3, 6, 24]. In the HCI field, biofeedback has increasingly been explored for its role in social interactions, such as sharing internal states to support emotional connection and by using inter-user physiological synchrony to assess relational dynamics [2, 33, 34]. Prior work utilized biofeedback for promoting mindful self-awareness [6], self-reflection and emotion regulation [9], empathy and compassion [16], and fostering social connections [23]. Social psychophysiology also investigated how synchronized physiological patterns can indicate emotional closeness or collaboration [36, 40, 45].

Most social biofeedback research develops hardware systems to evaluate the interactions within specific social contexts, such as smartphones [27], ambient installations [45], and VR headsets [22]. However, the developments often focused on a targeted experience, which leaves gaps for communicating and fabricating interdisciplinary explorations with low-fidelity prototypes. My research addresses the missing tools and processes for designing on-skin biofeedback interactions across disciplines. By proposing toolkits for grounding the discussion between HCI researchers and relevant fields' practitioners, I aim to examine how on-skin wearable systems, equipped with reconfigurable sensors and displays, can empower multidisciplinary practitioners in prototyping embodied designs of social biofeedback systems.

2.2 Design Strategies of Shared Biofeedback Systems

Although biofeedback can benefit interpersonal relationships by fostering intimacy, social connectedness, and emotional awareness, the use of biodata requires regulations to mitigate potential harm to individuals and society [33, 34, 39]. Recent research has discussed design strategies to address challenges involving environmental factors, Input/Output (I/O) designs, communication concepts, cognitive perceptions, and ethical considerations [39]. Hassib *et al.* [10] investigated users' interest in acquiring and sharing biofeedback, which showed user preferences in display modalities beyond the touchscreen. Liu *et al.* [26, 28] explored interactions of sharing biosignals in expressive ways. Feijt *et al.* [33] synthesized frameworks for conceptualizing biofeedback communication along dimensions like symmetry, synchronicity, and social context. Elagroudy *et al.* [5] utilized a BCI design probe to survey the design parameters for implicitly sharing cognitive states. Noura *et al.* [11] utilized a technology probe to investigate wearable biosignal displays, indicating that the ambiguous signals can result in mixed social cues, enhancing and complicating user perceptions.

Overall, these studies show the importance of regulating biodata visibility and meaning, aligning feedback modalities with user goals, and designing with ambiguity to preserve agency. Building on these design concerns, I will engage multidisciplinary experts through a co-design process, identifying strategies for balancing visibility, context, and communication intent.

2.3 Opportunities in On-skin Biofeedback Displays

Recent advances in soft materials and electronic technologies have made it possible to create miniature circuitry that can be comfortably worn and even directly applied to the skin, functioning as on-skin interfaces [18]. The micrometer-thickness devices can perform physiological sensing functions such as artery pulse [38, 41] and skin temperature [7], and support ultraflexible and conformable display using polymer light-emitting diodes (PLEDs) [47]. Recent HCI research builds on the vision of the *invisible computer* [44], extending ubiquitous computing to interface human skin to enable novel wearable interactions. In addition to exploring diverse sensing [15, 25, 30, 32, 42, 43] and display [8, 14, 20, 31, 43, 48] functions, prototyping technologies have been developed to democratize the creation of on-skin interfaces by using accessible materials and fabrication methods [19–21, 35].

Studies by Kao *et al.* [13] explored users' perception and comfort when wearing on-skin devices in public. Khan *et al.* [17] combined user-centered design preferences and engineering-centered biosignal data acquisition into a wear index to identify wear locations to inform designs of the on-skin stress monitoring device. A framework of skin interface aids in anticipating user needs and potential challenges [29], evaluating body locations, social acceptability, and wearability. These investigations highlight the need to integrate material and social considerations early in the design process. In response, I propose developing modular on-skin prototyping toolkits [19–21] to study the use of on-skin sensors and displays for health and social biofeedback applications.

3 methodology

The proposed research agenda is driven by the potential of on-skin biofeedback systems in enhancing social interactions and well-being through physiological data acquisition and embodied expression. To guide this exploration, I utilized a research-through-design [49] approach that unfolds across three stages:

- (1) Enable accessible on-skin prototyping using modular construction toolkits
- (2) Participatory Design with Multidisciplinary Practitioners
- (3) Develop context-aware social biofeedback systems

In this section, I first elaborate on how I approach the first two stages with the completed and ongoing research. Based on the findings, I describe the future work in the third stage and discuss the potential implementation of context-aware biofeedback designs.

3.1 Modular Toolkit Development

My research started with developing modular prototyping toolkits to democratize and facilitate the design of on-skin biofeedback interfaces. These toolkits address the unique challenges of integrating soft, stretchable electronics onto human skin by offering easy reconfigurability and low entry barriers.

3.1.1 SkinKit. The first step of the effort focuses on lowering the barrier to constructing on-skin prototypes. I introduced SkinKit [21], a modular prototyping platform allowing quick assembly and testing of on-skin electronic circuits, leveraging tessellated, skin-conformable substrates combined with flexible printed circuit board



Figure 1: SkinKit modules are made of slim, skin-conformable tessellated base substrates, with Flexible PCB blocks for wearing on diverse body locations.

(FPCB). SkinKit enables fast, low-fidelity prototyping with a slim form factor directly applicable to the skin. It leverages a variety of preprogrammed modules connected in unique sequences to achieve various function customizations, aiming to lower the barrier for building on-skin interfaces, as shown in Figure 1.

Through multi-session workshops with 34 maker participants spanning STEM and design backgrounds, the result revealed how diverse maker populations engage in on-skin interface design, what applications they chose to build, and what challenges they faced. To encourage broader participation, we collaborated with a local science center and the 4-H organization to promote STEM learning for middle school students using accessible on-skin technology [12]. The hands-on experience of prototyping on-skin circuitry can inspire participants and provide valuable insights into how and what users can build with SkinKit and uncover usability issues.

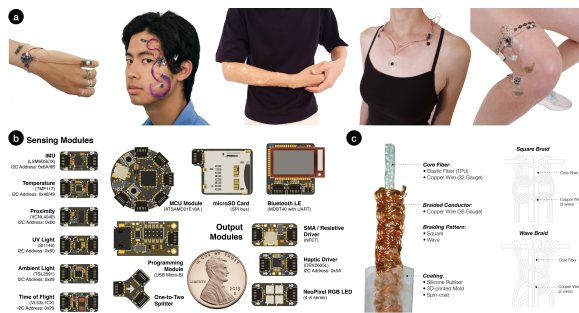


Figure 2: (a) SkinLink enables prototyping of on-skin interfaces through linking, adjusting, and deploying circuitry directly on the body. (b) Initial 14 circuit modules with sensing, computing, actuation, and communication functions. (c) SkinBoard adaptive trace module made with multi-ply braiding structures.

3.1.2 SkinLink. Since the overhead cost for on-skin prototyping comes from the device specificity that only serves one body location and one purpose, a toolkit approach that builds **out-of-kit** allows reconfigurable prototypes for design explorations. When a prototype fails to fit the desired body location, enabling **in-situ**

adjustments directly on the body would make the device adaptable without needing to repeat the prototyping process from scratch. Based on this idea, I proposed SkinLink [19], the second iteration of the on-skin toolkit that enables on-body prototyping through an adaptive and adjustable wiring process for out-of-kit on-skin interface fabrication. The reconfigurable on-skin fabrication approach allows users to intuitively explore and experiment with the circuitry adjustment on the body. SkinLink comprises tiny distributed circuit modules and a variety of streamlined trace modules to address the design challenge of the occasional stretches to the circuit due to body movements (Figure 2).

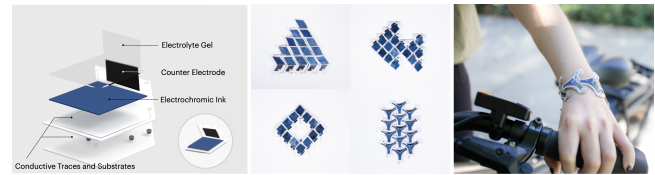


Figure 3: ECSkin is a construction toolkit for fabricating on-skin electrochromic displays. An ECSkin display is flexible and soft, easily conforming to curvy body parts.

3.1.3 ECSkin. Additionally, I introduced ECSkin [20] as a toolkit for fabricating modular, reconfigurable electrochromic displays, enhancing the on-skin prototypes with the subtle, flexible, and accessible output modality (Figure 3). Emerging electrochromic (EC) materials have advanced thin-film, low-power, and non-emissive display technologies. While suitable for wearable or on-skin applications, current EC display fabrication techniques are limited by size, response time, and manufacturing overhead when printing new patterns. The tessellated design of ECSkin complements the analog printed patterns and enables customizing the display area by combining pre-made modules.

3.2 Participatory Design with Multidisciplinary Practitioners

Utilizing the on-skin toolkits and display modalities, I will conduct participatory co-design workshops with multidisciplinary experts to explore the design space of on-skin biofeedback interfaces. The prototyping toolkit materials include sensors for heart rate variability (HRV) and galvanic skin response (GSR), along with visual, haptic, and electrochromic displays built on prior work [19, 20]. The participatory design process will enable experts to iteratively conceptualize, prototype, and refine interaction scenarios grounded in their disciplinary expertise. This process consists of structured brainstorming, hands-on prototyping, and reflective evaluation phases, enabling practitioners to experience making with the toolkit and interaction design. Through the interdisciplinary collaboration, I aim to identify design opportunities and challenges specific to on-skin social biofeedback interactions.

3.3 Context-Aware System Development

Another future work aims to integrate context-aware computational techniques with physiological sensing to advance the practicality and adaptability of on-skin interfaces to advance the practicality

and adaptability of on-skin interfaces. The proposed AI-supported biofeedback system should dynamically adapt biofeedback displays to reflect social context and user-defined preferences. The system will process physiological data in real-time, and high-level context interpretation will be performed by integrating multimodal sensing, *e.g.*, physiological and environmental sensors, with large language models (LLMs) for context-aware interaction. The AI agent should follow personalized preferences in guiding biofeedback visibility and expression, ensuring physiological disclosures align comfortably with the user's intent and social appropriateness. This approach combines physiological computing, modular on-skin interfaces, and advanced AI-driven reasoning, forming a comprehensive framework to explore situated, contextually sensitive interactions.

4 evaluation

4.1 Probing Social Contexts of Biofeedback

To investigate how users perceive and navigate on-skin biofeedback in diverse settings, I will conduct a design probe study by creating contextual probes with the on-skin prototyping platform. These probes will be applied to lab-simulated communication scenarios, such as private reflection, workplace interactions, formal interviews, and intimate conversations, to explore the role of biofeedback in modulating emotional expression, privacy, and social signaling. By engaging a broad audience of participants with the design probes in different scenarios, I will collect and analyze participants' reflections through survey questionnaires and semi-structured interviews. These studies aim to surface how users interpret, appropriate, or reject physiological feedback across social-relational contexts. By examining the significance of privacy, autonomy, interpretability, and the on-skin wearability, this work will inform how future biofeedback systems adapt responsively to user needs in complex social environments and provide design guidelines for improving user comfort and agency.

4.2 Context-Aware On-Skin Biofeedback Evaluation

To validate the feasibility of our proposed context-aware on-skin system, I will leverage the modular biofeedback prototypes developed from the SocialSkin project to investigate real-time identification of social contexts and adaptive biofeedback visualization aligned with user preferences and dynamic social cues. Real-time biosignal data will be collected and interpreted by leveraging multimodal sensing and in-context learning to detect social context and control biofeedback displays accordingly. The technical evaluation will assess the accuracy and reliability of the AI-driven system's context detection and adaptive responses.

- (1) Sensor Data Processing: Evaluation of the system's ability to reliably process and extract features from the physiological data streams (*e.g.*, HRV, GSR) in real-time, representing users' emotional and physiological states.
- (2) Context Detection Accuracy: Validating system predictions with user-annotated scenarios to quantify accuracy in recognizing conversational contexts and emotional dynamics.

- (3) AI Decision Fidelity: Assessing the alignment between AI-generated biofeedback display suggestions, user preferences, and qualitative user feedback.

Qualitative methods will be employed to assess system responsiveness, interpretability, and perceived appropriateness of feedback provided in different scenarios. Mixed-methods analysis, integrating thematic qualitative coding and quantitative data, will provide comprehensive insights into user perceptions, enabling refinement of design guidelines and ensuring that the system aligns closely with real-world interpersonal dynamics.

5 contribution

Through this positioning thesis, the proposed research contributes toward the development of context-aware on-skin biofeedback systems by integrating accessible prototyping, interdisciplinary co-design, and socially adaptive interaction design. I have introduced innovative modular toolkits that significantly lower barriers to creating on-skin interfaces. These toolkits enable multidisciplinary practitioners to explore, design, and prototype novel physiological interactions directly on the body, expanding the accessibility and flexibility of on-skin computing. In ongoing research, I deploy a participatory approach for engaging practitioners from diverse disciplines in the co-design of biofeedback systems. This participatory methodology provides rich, practice-informed insights into the nuanced social dynamics of biofeedback interactions, informing both system design and theoretical understanding. To enhance the inadaptive biofeedback, I will contribute a novel integration of physiological sensing and AI-supported context-awareness. I plan to develop systems capable of adapting biofeedback displays according to identified social contexts and individual user preferences to enhance the practicality and social appropriateness of the on-skin biofeedback interactions.

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